



Fast STEM image simulation in low-energy transmission electron microscopy by the accurate Chen-van-Dyck multislice method

G.S. Chen^a, Y.T. He^{b,c}, W.Q. Ming^{b,c,*}, C.L. Wu^a, D. Van Dyck^d, J.H. Chen^{b,c,*}

^a Center for High-Resolution Electron Microscopy, College of Materials Science & Engineering, Hunan University, Changsha, Hunan 410082, China

^b Pico Electron Microscopy Center, Innovation Institute for Ocean Materials Characterization Technology, Center for Advanced Studies in Precision Instruments, Hainan University, Haikou, Hainan 570228, China

^c Key Laboratory of Pico Electron Microscopy of Hainan Province, Hainan University, Haikou, Hainan 570228, China

^d EMAT, Department of Physics, University of Antwerp, Antwerpen 2020, Belgium

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ABSTRACT

The Chen-van-Dyck (CVD) formulation as a rigorous numerical solution to the Schrödinger equation has been demonstrated being the only accurate multislice method for calculating diffraction and imaging in low-energy transmission electron microscopy. The CVD formulation not only considers the forward scattering effects but also includes the backscattering effects. However, since its numerical computation has to be performed in real-space, the CVD method may suffer from divergence and inefficiency in computing time, especially when used for low-energy scanning transmission electron microscopy (STEM) image simulation. The present study investigates the influence of cutoff value and slice thickness on the accuracy and efficiency of STEM image simulation using this formula. The results show that a small cutoff value is required in the low-energy regime to ensure accuracy, especially for thick specimens. The optimal slice thickness can be predicted approximately by a simple equation. To speed up STEM imaging simulation by up to 17 times using the CVD formulation, a hybrid computation model incorporating multiple graphic process units (GPUs) is suggested, which is of great significance for quantitative STEM imaging in low-energy transmission electron microscopy. The significance of including backscattering effect in STEM image simulation is estimated in comparison with forward-scattering effect.

1. Introduction

Low-energy transmission electron microscopy (LE-TEM) has been demonstrated to be an effective way to reduce the knock-on damage for many irradiation-sensitive materials (Alem et al., 2009; Kaiser et al., 2011; Sasaki et al., 2010; Warner et al., 2010), such as carbon nanotubes, graphene, and so on. Therefore, developing advanced low-energy transmission electron microscopes with uncompromised spatial resolution has been a hot subject in the literature (Lee et al., 2012; Linck et al., 2016; Sasaki et al., 2010; Sawada et al., 2015), although it suffers from degraded spatial resolution due to chromatic and geometric aberrations (Sasaki et al., 2010). Up to date, a resolution of 0.14 nm has been achieved at 15 kV, making it possible to image graphene at atomic resolution with negligible damage (Sasaki et al., 2014). To quantitatively analyze the contrast and absolute values of these images at such a low voltage, an accurate image simulation method is essential.

The two popular numerical methods for solving the Schrödinger equation in electron microscopy are the Bloch and multislice (MS) methods. Due to its convenience in addressing the periodicity problem of defective crystals and its speed in numerical computation, the multislice method has been widely applied and investigated. However, in the conventional multislice (CMS) methods, the incident electrons are supposed to satisfy the high-energy approximation (≥ 100 keV) (Bird, 1989; Chen and Van Dyck, 1997; Cowley and Moodie, 1957; Goodman and Moodie, 1974), which ignores a second-order partial differential term. In addition, the parabolic wave front of the electron wave introduced by the Fresnel propagator deviates from spherical propagation. A simple but effective correction to this problem is to replace the Fresnel propagator with a spherical propagator from a mathematical point of view (Bird, 1989; Kilaas et al., 1987) and was named as the propagator-corrected multislice (PCMS) (Ming and Chen, 2013). It was shown that, to some extent, the error introduced by the high-energy

* Corresponding authors at: Pico Electron Microscopy Center, Innovation Institute for Ocean Materials Characterization Technology, Center for Advanced Studies in Precision Instruments, Hainan University, Haikou, Hainan 570228, China.

E-mail addresses: wqming@hainanu.edu.cn (W.Q. Ming), jhchen123@hainanu.edu.cn (J.H. Chen).

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approximation can be compensated for by this method when the accelerating voltage is higher than 50 kV. However, with lowering the voltage further, the PCMS method will also introduce significant errors, especially for high-angle electron scattering or thick specimens (Ming and Chen, 2013).

In 1997, without assuming the high-energy approximation (Chen and Van Dyck, 1997), Chen and van Dyck derived a rigorous solution to the relativistically-corrected Schrödinger equation for calculating electron diffraction and imaging within a multislice scheme, including both the forward and backward electron scattering effect. Based on these formula, several numerical investigations and tests have been carried out for image and diffraction simulation using forward scattering in TEM (Cai and Chen, 2012; Lv et al., 2015; Ming and Chen, 2013; Wacker and Schroder, 2015), such as the revised real space (RRS) (Cai and Chen, 2012) and the fully-corrected multislice (FCMS) (Ming and Chen, 2013) methods. In 1976, Van Dyck (Van Dyck, 1976) discussed the importance of backscattering in high-energy electron diffraction calculations. In 2015, Spiegelberg et al. (Spiegelberg and Rusz, 2015) realized the numerical computation of the electron backscattering effect and found that backscattering has a significant influence on the high-angle scattering. But the influence of backscattering needs more investigation especially at low-energy regime. We refer to these approaches as the Chen-van-Dyck multislice (CVDMS) method or formula hereafter for consistency (Cai and Chen, 2012; Lv et al., 2015; Ming and Chen, 2013; Wacker and Schroder, 2015). The RRS considered only the first- and second-order terms in the Taylor expansion, whereas the CVDMS method actually requires a cutoff criterion to control the computation precision of all the Taylor expansion terms. It is important to note that for low-energy TEM (< 20 kV), the CVDMS method has to be employed, since the errors of the CMS and the PCMS methods increase with the electron wavelength (Ming and Chen, 2013). Despite its advantage, the CVDMS method has not been extensively used. The reasons are twofold. Firstly, the CVDMS formula can only be numerically computed in real space. It is known that the real space multislice algorithms suffer from a divergence problem due to improper selection of sampling interval and slice thickness (Self, 1982; Wacker and Schroder, 2015; Wang and Chen, 1988; Zhu et al., 1993). Its accuracy also depends on how to calculate the Laplace operator (Wacker and Schroder, 2015). Secondly, the CVDMS computation is time-consuming compared with the PCMS and the CMS algorithms. It was reported that, as compared with the first-order term in CMS, the second-order term in the CVDMS contains many oscillations along the beam direction within a small distance (Hillier et al., 2017). Therefore, the slice thickness should be sufficiently small (more slice numbers) to simulate this oscillation. Otherwise, a computation error may arise. In addition, the number of terms used in Taylor expansions in the CVDMS formula is controlled by a cutoff value, heavily influencing both accuracy and computation speed. To overcome these difficulties, the influences of these factors on the accuracy and the efficiency of the CVDMS method need to be investigated thoroughly.

Moreover, the wide application of STEM techniques, including high-angle annular dark field (HAADF), annular bright field (ABF), ptychography, integrated differential phase contrast (iDPC), and so on, for solving crystal structures and atom arrangements at sub-angstrom resolution (Bosch and Lazić, 2015; Chen et al., 2021; Lazić et al., 2016; Yücelen et al., 2018), or even to extracting three-dimensional information from STEM images (Blom and Vogt, 2020; Chang et al., 2020; Ishikawa et al., 2014; Lu et al., 2015), has put forward higher requirements for accurate and efficient image simulation. However, STEM image simulation takes a lot more time than TEM image simulation because an entire multislice iteration must be performed for each probe position (pixel in STEM image). Although several studies on accelerating CMS/PCMS simulations of STEM images using graphic processing unit (GPU) have been reported (Lobato and Van Dyck, 2015; Pryor et al., 2017; Radek et al., 2018; Yao et al., 2016), no similar study has been carried out for the CVDMS simulations.

The present study focuses on investigating the influence of voltage, slice thickness, and cutoff value on the accuracy and efficiency of the CVDMS method, including the backscattering effects. It attempts to present a fast and accurate CVDMS algorithm using heterogeneous programming with multiple GPUs. In Section 2, the CVDMS formula is introduced briefly. Details of this formula can be found in Ref. (Chen and Van Dyck, 1997). In Section 3, the accuracy and speed of the CVDMS method at various voltages are investigated for STEM image simulation first. Then, a computation model using heterogeneous programming is presented. The influences of slice thickness and cutoff value are also discussed in detail, using a simple formulation to predict approximately the optimal slice thickness. Finally, the difference between the CVDMS methods with and without including backscattering effect is estimated.

2. Computational formula

The Dirac equation was derived based on the relativistic quantum mechanics (Dirac, 1927). Therefore, it can be used to describe the wave function of high-speed fermions considering spin effect, such as high-energy electrons. When the velocity of electrons is significantly smaller than that of light and the spin effect can be neglected, this equation can be reduced to the Schrödinger equation. More specifically, for a TEM with moderate accelerating voltage or lower, the interaction between the incident electrons and specimen, or wave function $\Psi(\mathbf{r})$, can be described by the “time-independent relativistically corrected” Schrödinger equation with sufficient accuracy (Fujiwara, 1961):

$$\left[\frac{\hbar^2}{8\pi^2 m} \nabla_r^2 + eU(\mathbf{r}) + \frac{\hbar^2 K_0^2}{2m} \right] \Psi(\mathbf{r}) = 0, \quad (1)$$

where m and K_0 are the relativistically corrected mass and the wave vector of the incident electron, respectively. $\hbar$ is the Planck constant. $\Delta = \nabla_r^2$ is the three-dimensional (3D) Laplace operator. $\mathbf{r} = (x, y, z)$ is the 3D real-space coordinate. e is the electron charge, and $U(\mathbf{r})$ is the specimen potential.

To solve this equation, several methods have been proposed, such as the Bloch wave methods (Van Dyck, 1975) and the multislice methods (Cowley and Moodie, 1957). Here, only the multislice will be discussed. By separating the wave function into two factors $\Psi(\mathbf{R}, z) = \varphi(\mathbf{R}, z) \exp(2\pi i K_0 z)$ and substituting it into Eq. (1), we have:

$$\left[-\frac{i\Delta_{xy}}{4\pi K_0} + \frac{\partial}{\partial z} + \frac{1}{4\pi K_0} \frac{\partial^2}{\partial z^2} - i\sigma U(\mathbf{r}) \right] \varphi(\mathbf{R}, z) = 0 \quad (2)$$

Where $\mathbf{R} = (x, y)$ is the 2D real-space coordinate, $\Delta_{xy} = \partial^2/\partial x^2 + \partial^2/\partial y^2$ is the 2D Laplace operator and $\sigma = 2\pi em/(\hbar^2 K_0)$ is the interaction constant. The detailed calculation formulas of the $U(\mathbf{r})$ can be found in Eqs. (11)–(14) of reference (Chen et al., 1997). The atomic scattering factors are calculated using the method introduced in Kirkland's book (Kirkland, 1998).

In the conventional multislice (CMS), the second order terms in this equation are neglected for convenience, thus introducing the high energy approximation.

$$\frac{\partial \varphi(\mathbf{R}, z)}{\partial z} = i \left[\frac{\Delta_{xy}}{4\pi K_0} + \sigma U(\mathbf{R}, z) \right] \varphi(\mathbf{R}, z) \quad (3)$$

However, the high energy approximation is not suitable for the LE-TEM simulation. Chen and Van Dyck proposed a rigorous multislice method to solve Eq. (1) without introducing any high energy approximation. Details about this can be found in Ref. (Chen and Van Dyck, 1997). Briefly return to the $\Psi(\mathbf{r})$, Eq. (1) can be rewritten as:

$$\frac{\partial^2 \Psi(\mathbf{r})}{\partial z^2} + [2\pi \hat{k}(\mathbf{r})]^2 \Psi(\mathbf{r}) = 0 \quad (4)$$

Where

$$\hat{k}(\mathbf{r}) = \pm K_0 \sqrt{1 + \frac{1}{(2\pi K_0)^2} \Delta_{xy} + \frac{\sigma}{\pi K_0} U(\mathbf{R}, z)} \quad (5)$$

The positive and the negative wave-vector operators $\hat{k}(\mathbf{r})$ indicate respectively the forward-scattered wave and the back-scattered wave. As described previously, $\Psi(\mathbf{R}, z) = \varphi(\mathbf{R}, z) \exp(2\pi i K_0 z)$. In the low-energy regime, backscattering effects should be considered. If considering only the so-called “single backscattering approximation”, under the framework of the multislice method, in which the specimen is considered as a combination of a series of slices with thickness ε perpendicular to the z -direction, the wave function at depth $z = j\varepsilon$ can be calculated from the following iterative formulation:

$$\varphi(\mathbf{R}, j\varepsilon) = (1 - \hat{B}_{j+1, j}) \exp[\hat{H}(\mathbf{R}, j\varepsilon)] \varphi[\mathbf{R}, (j-1)\varepsilon] \quad (6)$$

Where the back-scattering-coefficient operator is

$$\hat{B}_{j+1, j} = \frac{\hat{k}_{j+1} - \hat{k}_j}{2\hat{k}_{j+1}} = \frac{\hat{k}[\mathbf{R}, (j+1)\varepsilon] - \hat{k}(\mathbf{R}, j\varepsilon)}{2\hat{k}[\mathbf{R}, (j+1)\varepsilon]} \quad (7)$$

and

$$\hat{H}(\mathbf{R}, j\varepsilon) = 2\pi i \varepsilon K_0 \left[\sqrt{1 + \frac{\Delta_{xy}}{(2\pi K_0)^2} + \frac{\sigma U(\mathbf{R}, j\varepsilon)}{\pi K_0}} - 1 \right]. \quad (8)$$

Therefore, the wave-function at $z = n\varepsilon$ is

$$\varphi(\mathbf{R}, n\varepsilon) = \left\{ \prod_{j=1}^n (1 - \hat{B}_{j+1, j}) \exp[\hat{H}(\mathbf{R}, j\varepsilon)] \right\} \varphi(\mathbf{R}, 0) \quad (9)$$

To date, numerical computation of Eq. (9) can only be performed using real-space algorithms (Cai and Chen, 2012; Ming and Chen, 2013). Although the computational complexity of data processing in the real-space method (N^2) is theoretically smaller than that of the Fast Fourier Transform (FFT), which is $N^2 \log(N^2)$, where N is the number of a dimension, it actually takes more time in practice because the sampling interval of the real-space method needs to be smaller. In the CVDMS, the situation becomes worse because of the square root and exponent terms in Eqs. (7–9) must be calculated using the Taylor expansions, which takes a large amount of time. Eqs. (10–13) are Taylor expansions of the exponent term when propagating the electron wave from the $(j-1)^{\text{th}}$ slice to the j^{th} slice and the square root term. Many high-order terms need to be calculated iteratively to ensure accuracy. A critical question here is how many terms should be included in the Taylor expansion. In our previous study (Ming and Chen, 2013), we defined a criterion, or cutoff value. The $(k+1)$ -order iteration shall stop if the obtained intensities for the k -order terms over all pixels are smaller than the cutoff value.

$$\varphi(\mathbf{R}, j\varepsilon) = (1 - \hat{B}_{j+1, j}) \exp[\hat{H}(\mathbf{R}, j\varepsilon)] \varphi[\mathbf{R}, (j-1)\varepsilon] = \left(1 - \frac{\hat{k}[\mathbf{R}, (j+1)\varepsilon] - \hat{k}(\mathbf{R}, j\varepsilon)}{2\hat{k}[\mathbf{R}, (j+1)\varepsilon]} \right) \left[1 + \hat{H}(\mathbf{R}, j\varepsilon) + \frac{1}{2!} \hat{H}(\mathbf{R}, j\varepsilon)^2 + \frac{1}{3!} \hat{H}(\mathbf{R}, j\varepsilon)^3 + \dots \right] \varphi[\mathbf{R}, (j-1)\varepsilon] \quad (10)$$

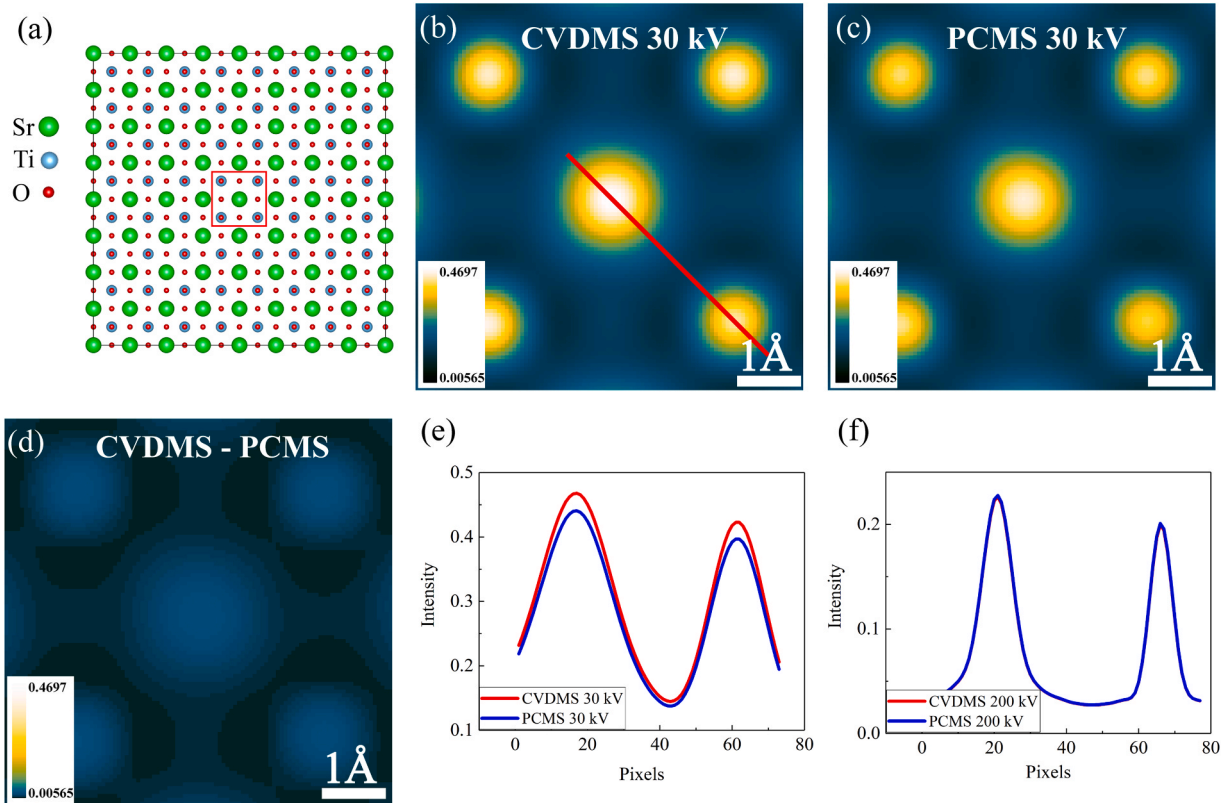


Fig. 1. HAADF STEM images of SrTiO_3 (a) simulated by the CVDMS (b) and PCMS (c) at 30 kV. (d) Difference of (b) and (c). (e) and (f) Intensity profiles along the solid yellow line in (b) and the same positions in (c) under 30 kV, as well as in images simulated by the PCMS and CVDMS method under 200 kV. The slice thickness is $0.2192 \text{ \AA}$ and cutoff value is 1×10^{-10} .

$$\hat{H}(\mathbf{R}, j\epsilon) = 2\pi i \epsilon K_0 \left\{ \frac{1}{2\pi K_0} \left[\frac{\Delta_{xy}}{4\pi K_0} + \sigma U(\mathbf{R}, j\epsilon) \right] - \frac{1}{8(\pi K_0)^2} \left[\frac{\Delta_{xy}}{4\pi K_0} + \sigma U(\mathbf{R}, j\epsilon) \right]^2 + \dots \right\} \quad (11)$$

$$\hat{k}(\mathbf{R}, j\epsilon) = K_0 \left\{ 1 + \frac{1}{2\pi K_0} \left[\frac{\Delta_{xy}}{4\pi K_0} + \sigma U(\mathbf{R}, j\epsilon) \right] - \frac{1}{8(\pi K_0)^2} \left[\frac{\Delta_{xy}}{4\pi K_0} + \sigma U(\mathbf{R}, j\epsilon) \right]^2 + \dots \right\} \quad (12)$$

$$\frac{1}{\hat{k}(\mathbf{R}, j\epsilon)} = \frac{1}{K_0} \left\{ 1 - \frac{1}{2\pi K_0} \left[\frac{\Delta_{xy}}{4\pi K_0} + \sigma U(\mathbf{R}, j\epsilon) \right] + \frac{3}{8(\pi K_0)^2} \left[\frac{\Delta_{xy}}{4\pi K_0} + \sigma U(\mathbf{R}, j\epsilon) \right]^2 + \dots \right\} \quad (13)$$

To ensure accuracy of the Laplace operator, a 9-point method was incorporated in the CVDMS method (Wacker and Schroder, 2015). The 9-point method is presented as following expressions. For the x and y components of Δ_{xy} :

$$\frac{\partial^2}{\partial x^2} \varphi(\mathbf{R}, j\epsilon) = \frac{\partial^2}{\partial x^2} \varphi(x, y, j\epsilon) = \frac{1}{\Delta x^2} \left(\frac{8}{5} \eta_1 - \frac{1}{5} \eta_2 + \frac{8}{315} \eta_3 - \frac{1}{560} \eta_4 \right) \quad (14)$$

$$\frac{\partial^2}{\partial y^2} \varphi(\mathbf{R}, j\epsilon) = \frac{\partial^2}{\partial y^2} \varphi(x, y, j\epsilon) = \frac{1}{\Delta y^2} \left(\frac{8}{5} \zeta_1 - \frac{1}{5} \zeta_2 + \frac{8}{315} \zeta_3 - \frac{1}{560} \zeta_4 \right) \quad (15)$$

where Δx and Δy are the sampling interval, $\eta_p = \varphi(x+p, y, j\epsilon) + \varphi(x-p, y, j\epsilon) - \varphi(x, y, j\epsilon)$, $\zeta_p = \varphi(x, y+p, j\epsilon) + \varphi(x, y-p, j\epsilon) - \varphi(x, y, j\epsilon)$, and $p = 1, 2, 3, 4$.

Eqs. (5–15) denote the CVDMS method. It is seen from Eq. (11) that the propagator interacts with the phase grating when the high-order terms are included. If only the first-order term is considered and neglecting the backscattering effects, the equation is reduced to the CMS method, in which the propagator is obtained by a parabolic approximation. To correct the difference between the parabolic surface and the spherical wave front, a spherical propagator is introduced as Eq. (16). The multislice solution of Eq. (17) is named as the PCMS method without including the backscattering effects in (Bird, 1989; Kilaas et al., 1987; Ming and Chen, 2013).

$$\frac{\partial \varphi(\mathbf{R}, z)}{\partial z} = \left\{ 2\pi i K_0 \left[\sqrt{1 + \frac{1}{(2\pi K_0)^2 \Delta_{xy}}} - 1 \right] + i\sigma U(\mathbf{R}, z) \right\} \varphi(\mathbf{R}, z). \quad (16)$$

$$\varphi(\mathbf{R}, j\epsilon) = \exp \left\{ 2\pi i \epsilon K_0 \left[\sqrt{1 + \frac{1}{(2\pi K_0)^2 \Delta_{xy}}} - 1 \right] + i\epsilon \sigma U(\mathbf{R}, j\epsilon) \right\} \varphi[\mathbf{R}, (j-1)\epsilon] \quad (17)$$

Here, we present the formulas for the probe. The wave function $\varphi_p(\mathbf{R}, \mathbf{R}_p)$ of the focused probe incident upon the specimen at position $\mathbf{R}_p$ is formed by integrating the aberration wave function $\exp[-i\chi(\mathbf{q})]$ over the objective aperture as (Kirkland, 1998):

$$\varphi_p(\mathbf{R}, \mathbf{R}_p) = A_p \int_0^{q_{\max}} \exp[-i\chi(\mathbf{q}) - 2\pi i \mathbf{q} \cdot (\mathbf{R} - \mathbf{R}_p)] d^2 \mathbf{q} \quad (18)$$

Where $q_{\max} = \alpha_{\max}$ (λ , the wavelength of the incident electron) is the maximum angle in the objective aperture, $\mathbf{q} = (q_x, q_y)$ is 2D spatial frequency in the Fourier transform of the image plane and A_p is a normalization constant chosen to yield:

$$\int |\varphi_p(\mathbf{R}, \mathbf{R}_p)|^2 d^2 \mathbf{R} = 1 \quad (19)$$

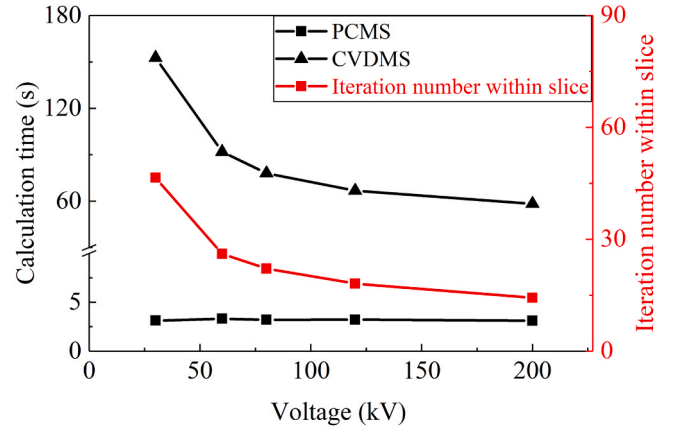


Fig. 2. The computation time of a single CBED simulation by the CVDMS and PCMS at different voltages on a CPU using 6 threads. The red curve shows the average iteration number within a slice for the CVDMS method at different voltages.

3. Results and discussion

3.1. Computation details

In the present study, a supercell consisting of $8 \times 8 \times 50$ unit cells of SrTiO_3 in three dimensions was constructed with the lattice constant being $31.561 \times 31.561 \times 197.257 \text{ \AA}$, as displayed in Fig. 1a. The sampling number of the potential and the probe were 512×512 , corresponding to pixel sizes of $0.0616 \times 0.0616 \text{ \AA}$. Since the purpose of the present study is about the accuracy and efficiency of multislice computation, we set spherical aberration and defocus to 0, and a semi-convergence angle of 35 mrad was selected in all calculations, independent of the accelerating voltage. The angular range of the HAADF detector was set to 52–135 mrad at 200 kV and 120–377 mrad at 30 kV. The thermal diffuse scattering was considered using 32 frozen lattices in which the Debye-Waller factors of all elements at 300 K were from (Gao and Peng, 1999). To save computation time, only the central part of the supercell, as indicated by a red rectangle in Fig. 1a, was simulated. At each slice of the calculation, the transmission function and the wavefunction (Eq. (10)) are bandwidth limited up to 2/3 of their reciprocal frequency to prevent artifacts (Self et al., 1983).

The PCMS was realized using fast Fourier transformation (FFT), and the CVDMS was realized using real space method.

3.2. Computation results

3.2.1. Performance of the CVDMS method

Fig. 1b-d show a comparison between HAADF STEM images simulated by the PCMS and CVDMS at 30 kV and their difference (Fig. 1d). All images share the same intensity scale. Significant difference exists between the CVDMS and PCMS at 30 kV, indicating that the PCMS method may induce errors in quantitative STEM calculation. To show the energy dependence of simulation accuracy, line profiles along the Sr and Ti-O atom columns (as indicated by the solid red line in Fig. 1b) in simulated images at 30 and 200 kV are displayed in Fig. 1e and Fig. 1f.

The efficiency of the CVDMS method was also evaluated for practical application. Fig. 2 shows the average computation time of a single CBED calculation by the two methods at various voltages on a CPU using 6 threads. The computation time of the PCMS is independent of the accelerating voltage. However, the time consumed by the CVDMS increases drastically when the accelerating voltage is lowered. Even at 200 kV, the consumed time of the CVDMS is about 18 times longer than the PCMS method, as can be seen in Fig. 2. Such a big difference can be attributed to the distinctive calculation method, i.e., the PCMS was performed using the FFT, while the CVDMS can only be realized

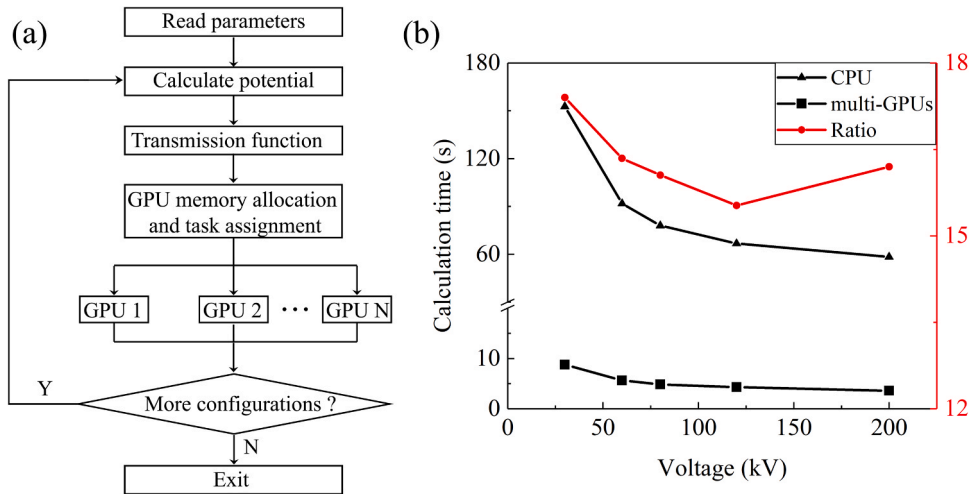


Fig. 3. (a) The flow chart of STEM image simulations. (b) The calculation time of single CBED pattern simulations by CVDMS on multi-GPUs and on CPU using 6 threads at different voltages, the red curve shows the ratio of calculation time between CPU and multi-GPUs.

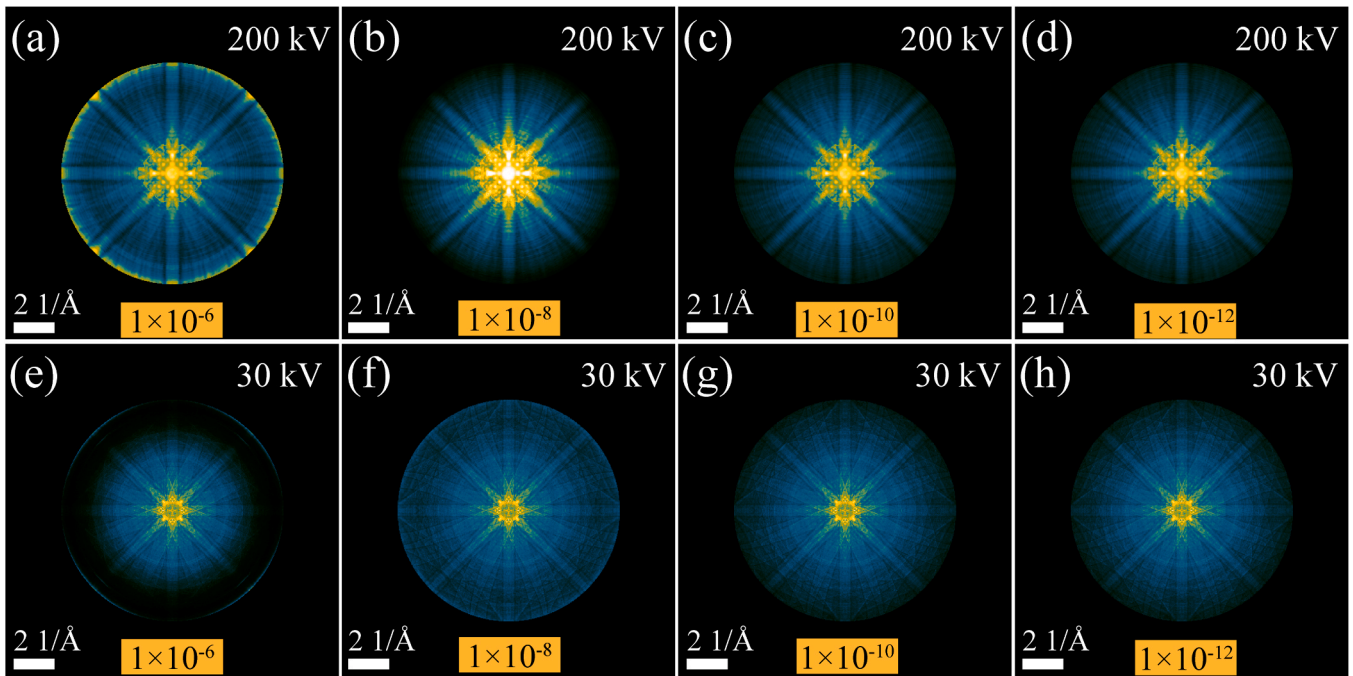


Fig. 4. The CBED patterns calculated using different cutoff values at 200 kV (a)-(d) and 30 kV (e)-(h). The cutoff values are labelled on corresponding images. The display ranges of (a-d) and (f-h) are the same. The slice thickness is $0.2192 \text{ \AA}$ at 30 kV and 200 kV. The CBED patterns were displayed using a logarithmic scale.

numerically in real space due to its mathematical complexity (Chen and Van Dyck, 1997). Under this scheme, the Laplace operator in the CVDMS was realized using a 9-point differential method to ensure accuracy (Wacker and Schroder, 2015). In addition, the square root and exponent terms in Eqs. (7–9) must be calculated using the Taylor expansions, which takes a large amount of time. These two terms are realized using two nested iterations in Eqs. (8–9) and three iterations in Eq. (7) in the CVDMS method. The average iteration numbers within a single slice are also shown in Fig. 2. The average iteration number increases simultaneously with the computation time, which indicates that the iteration number is the key to improving the efficiency of the CVDMS.

Another effective method to accelerate the CVDMS calculation is heterogeneous programming using multiple GPUs. In a STEM image, the value of a pixel is integrated from a calculated CBED pattern according to the selected detector range. This means that a full multislice

calculation should be performed for each pixel (electron probe position). The calculation at different pixel positions is independent of others and can, in principle, be performed simultaneously. Fig. 3a shows the flowchart of heterogeneous programming using multiple GPUs. The program is written in CUDA C++ . Even though it has been investigated in several papers (Lobato and Van Dyck, 2015; Pryor et al., 2017; Radek et al., 2018; Yao et al., 2016), such a programming method has not been applied to accelerate the CVDMS method.

As shown in Fig. 3a, once all parameters are input into the memory of a computer, the task (pixels of the STEM image) will be distributed to N GPU devices. In such a way, the speed of the calculation can be improved a lot, depending on the number and ability of the GPU devices used. It should be noted that these devices are very cheap, and most investigators can afford them. In the present study, an i7-6700K CPU with 8 threads, and two Nvidia 1080 GPUs were used. In the present study,

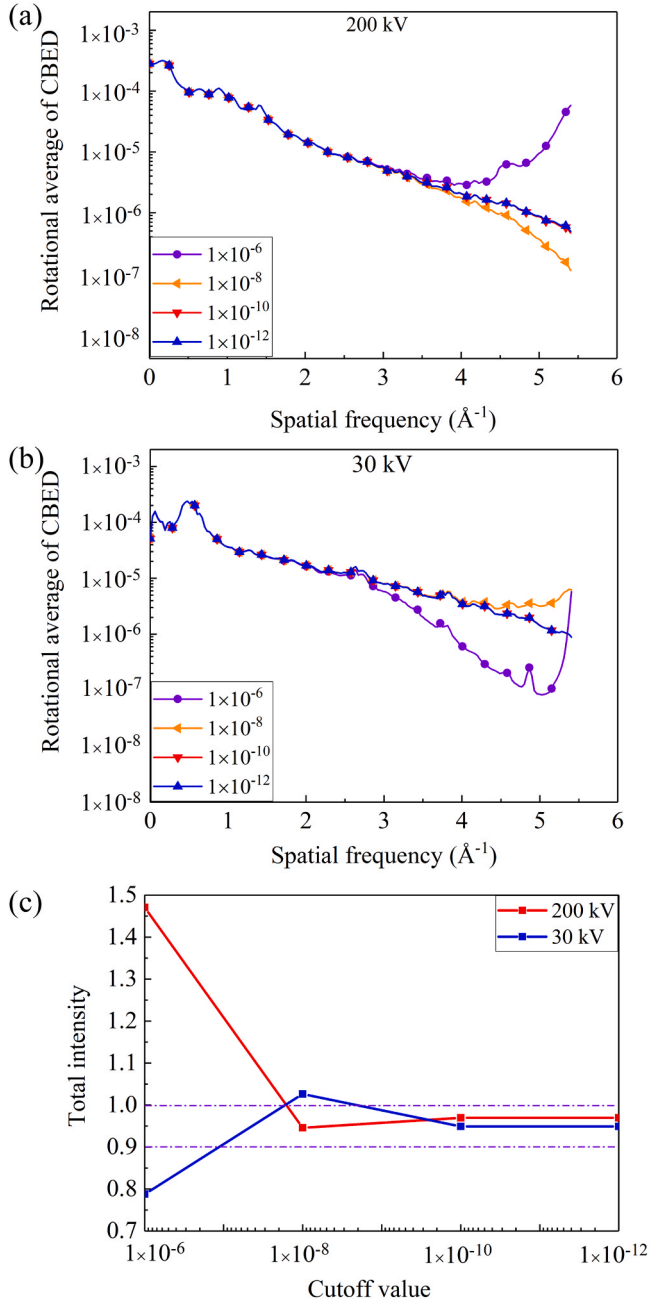


Fig. 5. The rotational average profiles of CBED patterns simulated at 200 kV (a) and 30 kV (b), respectively. (c) The total integrated intensity of the simulated CBED patterns plotted against various cutoff values. The slice thickness is 0.2192 Å at 30 kV and 200 kV. The rotational average profile of CBED patterns were displayed using a logarithmic scale.

the warp size is 32, and the block size is 128. The grid size was calculated according to the ratio between the array size and the block size. The average calculation time of a single pixel of a STEM image at various voltages is shown in Fig. 3b. As compared with CPU-based code using 6 threads, the acceleration ratio can be as large as 17, which is quite significant. It should be noted that the time consumed by a single pixel is less than 10 seconds when the voltage is higher than 30 kV, almost as fast as the CPU-based PCMS method (Fig. 2).

3.2.2. Optimization of parameters

Compared to the PCMS method, there are several parameters in the CVDMS method that need to be concerned with more attention, since

they heavily affect the accuracy and efficiency of the calculation. The first one is the cutoff value, which sets a criterion for when iteration can stop. This parameter may also influence the convergence of the multislice calculation, as seen in the following results. Another parameter that should be considered is slice thickness, as it can also lead to divergence and slow down the calculation.

Fig. 4 shows the influence of the cutoff value on CBED simulation using the CVDMS method at 200 kV and 30 kV. The electron probe was located at the center of a Sr atom column. For convenience, these two situations are renamed as the high-voltage case (200 kV) and the low-voltage case (30 kV), respectively. In principle, a large cutoff value promotes the calculation speed. However, it can also introduce errors. Comparing Fig. 4a-d visually, significant error can be seen at the high-angle range for the 200 kV situation when the cutoff value is 1×10^{-6} (Fig. 4a). When smaller cutoff values are used, the error disappears gradually (Fig. 4b-d). Similar phenomena can also be observed in the low-voltage case (Fig. 4e-h).

A more quantitative comparison of the rotational average profiles of Fig. 4a-h is displayed in Fig. 5a-b. Divergence occurs for both cases when the cutoff value is set to 1×10^{-6} and 1×10^{-8} . The profiles coincide with each other when the cutoff value further decreases to 1×10^{-10} and 1×10^{-12} . Furthermore, a normalization test was conducted using three different case cutoff values. The normalization test requires that the total integrated intensity of the simulated CBED patterns must be less than 1 and greater than 0.9. Fig. 5c shows the total intensity of the calculated CBED with different cutoff values. It is seen that with decreasing the cutoff value, 1×10^{-12} for example, the total intensity converges to a stable value. This is consistent with Figs. 4 and 5a-b.

Specimen thickness is a key parameter in image simulation. For a thin specimen, such as a weak phase object (WPOA), the high-angle scattering is very small, while for a thick specimen, the high-angle scattering becomes significant. Therefore, the influence of specimen thickness on the selection of the cutoff value should also be evaluated. Fig. 6 shows the difference in the rotational average profiles of a CBED simulated at 30 kV with two different cutoff values: 1×10^{-8} and 1×10^{-12} . The calculated profiles with a cutoff value of 1×10^{-12} are used as a criterion. The deviation from this profile means there are calculation errors. For a 32.876 Å thick specimen, the differences are nearly zero at all frequencies, indicating that the cutoff value of 1×10^{-8} is small enough in this case. With increasing specimen thickness, errors become larger.

From the above results, a small cutoff value should be chosen to perform accurate multislice simulation. However, a small cutoff value will also prolong the calculation time. Therefore, a proper selection of the cutoff value should be made for accurate and efficient simulation.

As described above, there are three layers of iterations in the real-space CVDMS method. The outermost is the iteration of the incident electrons passing through all slices. The other two layers of iteration are from the numerical realization of the exponent and square root terms in Eq. (7). The triple iterations of the CVDMS method are the key factor affecting its speed. Thus, besides using advanced heterogeneous programming techniques, reducing the number of iterations is another effective pathway to speed up calculations. It is obvious from Eqs. (7–9) that all three iterations are related to slice thickness. Therefore, it is meaningful to evaluate the influence of slice thickness on the CVDMS calculation. As is well known from previous studies (Chen and Van Dyck, 1997; Cowley and Moodie, 1957; Self et al., 1983), slice thickness should be quite small to ensure accurate multislice simulation. In general, a slice thickness of 2 Å or smaller is suggested (Self et al., 1983).

Fig. 7a shows the consumed computation time as a function of slice thickness. To our surprise, the total consumed time decreases with slice thickness drastically in the first stage. After a minimum is reached, the time cost increases inversely. This phenomenon demonstrates that there is an optimal value for slice thickness, with which the CVDMS can reach the highest calculation speed. Existence of an optimal slice thickness can be related to the total iteration number. As shown in Fig. 7a, the average

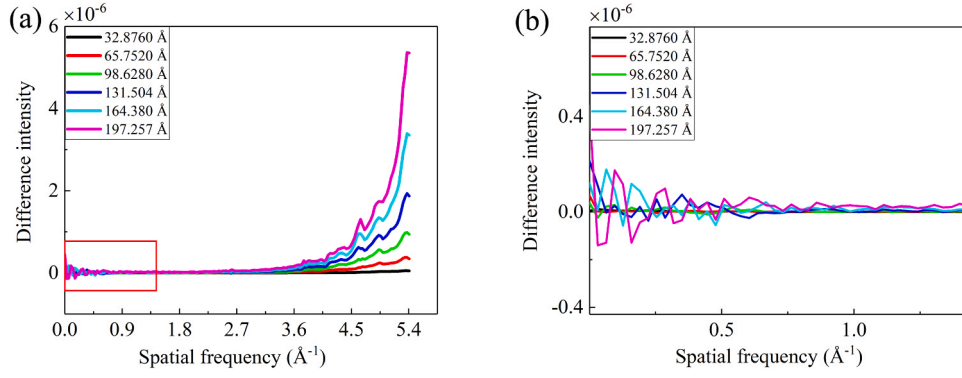


Fig. 6. The differences of the rotational average profiles of CBED simulated at 30 kV with different cutoff values: 1×10^{-8} and 1×10^{-12} . The specimen thickness varies from 32.876 Å to 197.257 Å. The slice thickness is 0.2192 Å.

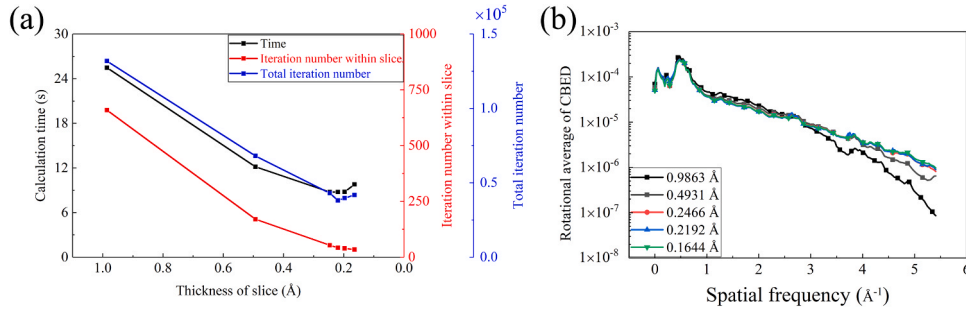


Fig. 7. (a) The relation between the consumed time and slice thickness. The average number of iterations within a slice and the total number of iterations are also shown. (b) The rotational average profile of CBED at different slice thicknesses, from 0.9863 Å to 0.1644 Å, for testing convergence. Calculations were performed at 30 kV. The rotational average profile of CBED patterns were displayed using a logarithmic scale.

iteration number within each slice decreases monotonically with reducing slice thickness, indicating that the calculation of $\hat{H}$ converges much faster when a smaller slice thickness is used. However, this cannot explain the relationship between time consumed and slice thickness. Actually, the total number of iterations, which is the sum of iteration numbers over all slices, will increase with reducing slice thickness because of the automatic increase in the total number of thinner slices used. Furthermore, an HAADF-STEM image simulation focuses more on the high spatial frequency signals, so CBED patterns that were used to integrate must be convergent. In Fig. 7b, different slice thicknesses were used to test the convergence. When the slice thickness is 0.9863 Å and 0.4931 Å, the line profile of the rotational average of CBED differs from the case with smaller slice thickness. Reducing slice thickness, when the slice thickness smaller than 0.2466 Å, the rotational average of CBED is

convergent. Therefore, the optimal slice thickness can ensure a high calculation speed and accuracy simultaneously.

Then, an important question is how to determine the optimal slice thickness using a simple method. In 2015, a stability criterion was proposed (Wacker and Schroder, 2015) based on the Courant-Friedrichs-Lewy (CFL) condition (Courant et al., 1928).

$$\varepsilon \leq 4K_0 \Delta x^2 C \quad (20)$$

where ε is slice thickness, Δx is sampling interval, and C is a real number depending on properties of the numerical algorithm, which can be set to 1 in the present case (D.W. Zingg, 2006). K_0 is the wave number of the incident electrons.

This equation was proposed to determine a proper slice thickness to fulfill the normalization test of the multislice method. To our surprise,

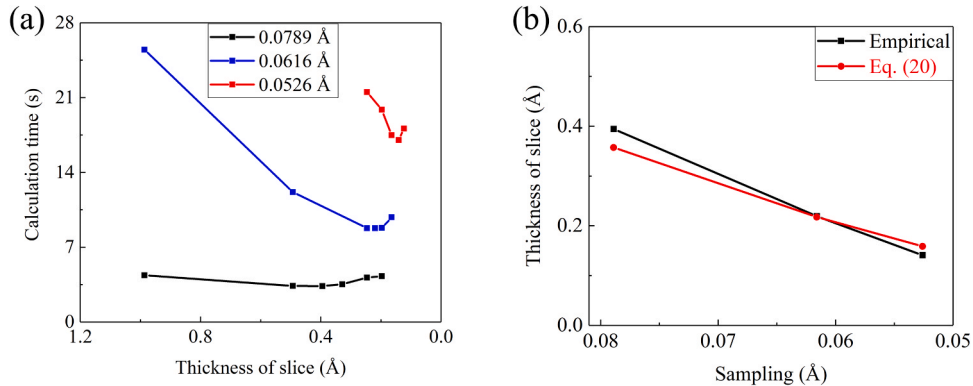


Fig. 8. (a) The relationship between the total time consumption and slice thicknesses when different sampling intervals were used. (b) Comparison between the optimal slice thickness values from (a) and those predicted by Eq. (20). All calculations were performed at 30 kV.

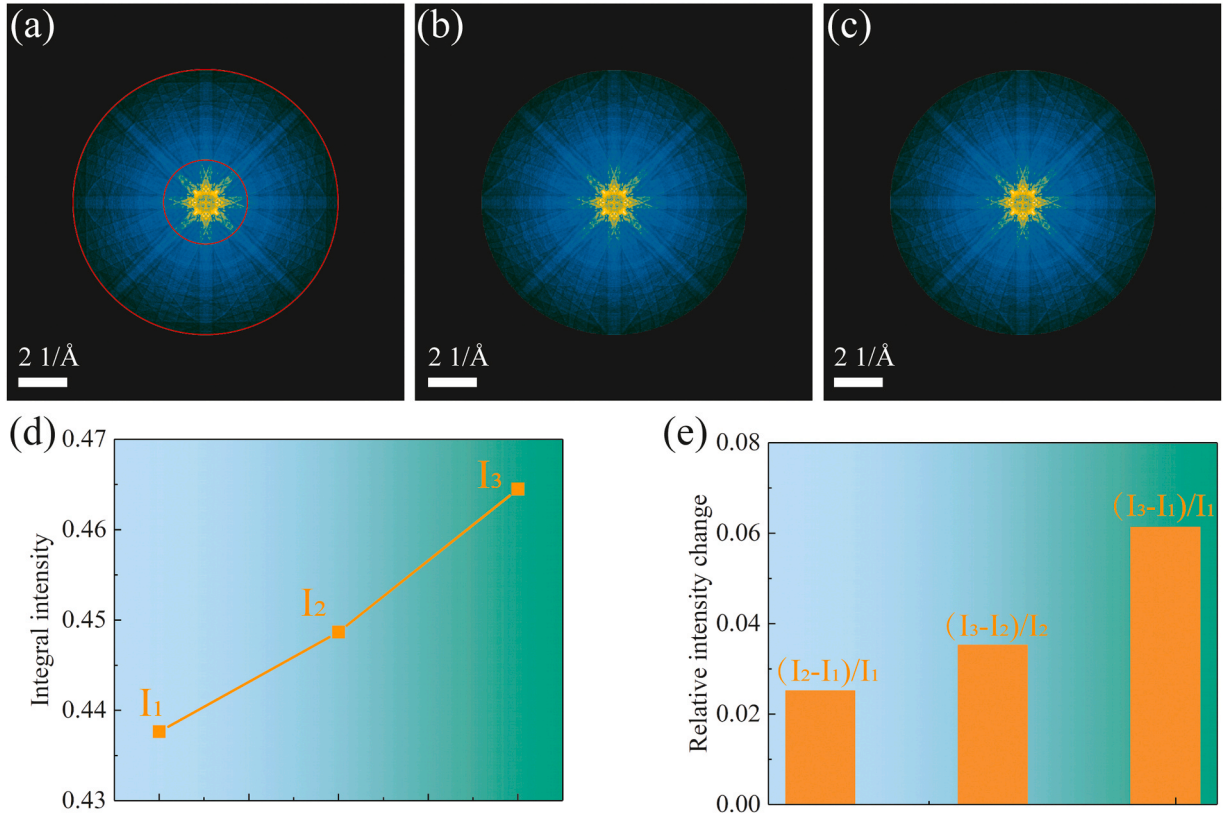


Fig. 9. The CBED patterns calculated at the Sr atom column by (a) PCMS at 30 kV, (b) CVDMS considering only forward scattering effects at 30 kV, (c) CVDMS considering both forward scattering effects and backscattering effects at 30 kV. (d) The integral intensities I_1 , I_2 and I_3 are calculated from (a) - (c), respectively. The integral angular range is 120–377 mrad, as marked by the two red circles in (a). (e) The relative intensities between I_2 and I_1 , I_3 and I_2 , I_3 and I_1 . The CBED patterns were displayed using a logarithmic scale.

we found that it can also be a perfect formula to predict the optimal slice thickness for the CVDMS calculation. Fig. 8a shows the computation time at various slice thicknesses when different sampling intervals were used. For all sampling intervals, a similar trend to that in Fig. 7a can be seen, where an optimal slice thickness can be found. Fig. 8b shows the optimal slice thickness from Fig. 8a plotted against the sampling interval. The values predicted by Eq. (20) were also plotted in this figure. The two curves almost coincide with each other. Therefore, Eq. (20) provides a simple way to predict the optimal slice thickness, which ensures a high calculation speed and accuracy simultaneously. As in most slice methods applied to crystals, such as this, there must be an integer number of slices in the unit cell. Thus, the number of slices in the unit-cell must be an integer, and it should approach the value predicted by Eq. (20).

3.3. Significance of backscattering effect

When considering only the forward scattering effect in the CVDMS formula, e.g., $B_{j+1j} = 0$, the forward scattering wave at $z = j\epsilon$ can be calculated from the following iterative formulation:

$$\varphi_{pf}(\mathbf{R}, j\epsilon) = \exp[\hat{H}(\mathbf{R}, j\epsilon)] \varphi(\mathbf{R}, (j-1)\epsilon) \quad (21)$$

Therefore, the wave-function at $z = n\epsilon$ is

$$\varphi_{pf}(\mathbf{R}, j\epsilon) = \prod_{j=1}^n \exp[\hat{H}(\mathbf{R}, j\epsilon)] \varphi(\mathbf{R}, 0) \quad (22)$$

In order to estimate the difference between the CVDMS methods with and without including backscattering effect, we will compare CBED patterns calculated by different formula and their related integral intensities at a Sr atom column. Fig. 9a-c show CBED patterns calculated respectively by the PCMS formula, the CVDMS formula considering only

forward scattering effects, and the CVDMS formula considering both forward and backscattering effects at 30 kV. All images share the same intensity scale. In Fig. 9d, the integral intensities I_1 , I_2 and I_3 are calculated respectively from Fig. 9a-c with the integral angle range from 120 to 377 mrad, as indicated by the two red circles in Fig. 9a which corresponds to a HAADF imaging detector ring. There are significant differences among the integral intensities I_1 , I_2 and I_3 . It is seen from Fig. 9e that for HAADF imaging in a low-voltage electron microscopy the backscattering effect can be quite significant. The difference between the two integral intensities calculated by the CVDMS and the PCMS formula can be up to 6 %, whereas that by the CVDMS formula with and without including backscattering effect can be up to 3.5 %. Fig. 9e demonstrates that not only the CVDMS method has to be used, but also the backscattering effect needs to be included for simulating HAADF-STEM imaging at low-voltages, e.g., at 30 kV or below.

4. Conclusions

In the present study, the accuracy and efficiency of the CVDMS method in low-energy TEM were evaluated by numerical simulations. Based on the results obtained, several concluding remarks can be drawn.

- (1) At low-voltages in HAADF-STEM imaging, the CVDMS method has to be used for accurate quantitative image simulations and also the backscattering effect needs to be included.
- (2) It requires more iterations for the CVDMS method to converge in the low voltage regime, thus reducing calculation efficiency. This difficulty can be overcome by heterogeneous parallel programming. The computation time of a CVDMS calculation executed on two GPUs simultaneously can be reduced by about 17 times in comparison with the codes run on CPU using 6 threads.

- (3) It is preferred to use a small slice thickness since it can reduce the computation time and avoid divergence simultaneously. The overall computation time first decreases with slice thickness. However, with further reducing the slice thickness, the computation time increases reversely due to the increase in the total iteration number. The optimal slice thickness can be well predicted by a simple formula.
- (4) To balance accuracy and efficiency of the CVDMS method, a proper cutoff value shall be used. A large cutoff value can decrease calculation time but may result in computational divergence. For image simulation of thick specimens, smaller cutoff values, e.g., 1×10^{-10} at 30 kV, should be used.

CRedit authorship contribution statement

Y.T. He: Writing – review & editing. **G.S. Chen:** Writing – original draft, Visualization, Validation, Software, Investigation, Data curation, Conceptualization. **C.L. Wu:** Writing – review & editing. **W.Q. Ming:** Writing – review & editing, Validation, Conceptualization. **Jianghua Chen:** Writing – review & editing, Supervision, Resources. **D. Van Dyck:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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